

Smart Electric Fence with Animal Detection System

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Abstract—Wild animals such as elephants, wild boars, deer, and other herbivores frequently enter inhabited and protected areas in search of food and water, resulting in property damage, threats to human life, and harm to wildlife. This paper presents an Intelligent Smart Electric Fencing System integrating an ESP32 microcontroller with an ESP32-CAM module, PIR sensors, and ultrasonic sensors for automated intrusion detection. The system activates relay-controlled electric pulse and visual alert mechanisms only upon confirmed detection, improving energy efficiency and minimizing unnecessary wildlife exposure to deterrents. IoT connectivity via Wi-Fi enables remote monitoring, real-time alerts, event logging, and cloud-based data management. Prototype testing demonstrated reliable animal detection, selective deterrence, and verified IoT dashboard functionality.

Keywords—smart electric fence, animal detection, ESP32, IoT, PIR sensor, wildlife intrusion, AI, human-wildlife conflict

I. INTRODUCTION

Human-wildlife conflict has become a major concern in regions where human settlements and forest boundaries exist in close proximity. Wild animals such as elephants, wild boars, deer, and other herbivores frequently move into inhabited and cultivated areas in search of food and water, leading to property damage, threats to human life, injuries, and fear among local communities [1]. Animals are simultaneously exposed to serious risks including injury, stress, habitat disturbance, and retaliatory harm from humans.

Traditional electric fencing systems operate continuously, consuming electrical power even when no threat is present [2]. This results in increased energy consumption, higher maintenance costs, and reduced system lifespan. Conventional systems also respond unnecessarily to environmental disturbances such as wind or vegetation movement, increasing risks for both humans and animals.

The proposed Intelligent Smart Electric Fencing System detects movement near protected boundaries and activates deterrent mechanisms only when required, improving energy efficiency and minimizing unnecessary exposure of wildlife to electric deterrents. Integration with IoT technology enables remote monitoring and real-time farmer alerts.

II. LITERATURE SURVEY

Panda et al. [1] proposed an IoT-based system using ultrasonic sensors and a camera module for wild animal intrusion detection with real-time notifications. The study demonstrated feasibility for crop protection but lacked automated deterrent mechanisms or intelligent fence control.

Kumar et al. [2] presented a smart monitoring system combining IoT sensors with neural-network-based animal detection for farmland protection. While improved automation was demonstrated, the system lacked physical deterrence or automated fencing. Mamat et al. [3] introduced a YOLOv5-based animal detection system achieving improved accuracy but without electric fencing integration.

Manikandan et al. [4] proposed an ESP32-based surveillance system but lacked automated deterrent actions and fence integrity monitoring. Daniel Raj et al. [5] presented an AI-based wildlife detection system for preventing vehicle-animal collisions, primarily transportation-oriented without addressing automated fencing.

Ji and Zhu [6] described an embedded YOLO-based recognition system focusing on computational efficiency without deterrence mechanisms. Ma and Yang [7] proposed an improved YOLOv5 network without IoT-based monitoring or automated deterrence. Priya et al. [8] developed a CNN-based species classification system without physical prevention systems.

Viji et al. [9] presented a wildlife detection framework lacking IoT integration and protective fencing. Giordano et al. [10] discussed LoRa/ZigBee architectures for monitoring without controlled fence response. Gandhi et al. [11] proposed deep-learning-based animal detection without physical protection. Knyva et al. [12] proposed a PIR/camera IoT system for road safety without multi-zone fence monitoring.

III. PROBLEM ANALYSIS

Existing wildlife intrusion management systems exhibit five critical limitations:

- (1) Continuous fence energization regardless of intrusion presence leads to unnecessary energy consumption.
- (2) Simple motion detection produces high false alarm rates from environmental disturbances.
- (3) Systems lack intelligent animal detection capability to differentiate wildlife from harmless activity.
- (4) No dynamic deterrence escalation exists to selectively activate protection only when required.
- (5) Absence of centralized monitoring, event logging, and remote communication reduces operational reliability.

In addition to these limitations, conventional fencing systems often require frequent maintenance due to continuous operation and prolonged energization of fence components. Constant electrical activation reduces the operational lifespan of relays, drivers, and power circuitry, increasing maintenance cost and reducing overall system efficiency. Many traditional systems also depend heavily on manual supervision, making them less suitable for deployment in remote agricultural regions where continuous human monitoring is impractical.

Environmental conditions further affect the reliability of existing intrusion detection systems. Weather conditions such as rain, fog, strong winds, and low-light environments can reduce sensing accuracy and increase false triggering events. Motion-only detection systems are unable to perform reliable intrusion verification, resulting in unnecessary activation of deterrent mechanisms even when no real wildlife threat exists. This inefficient operation leads to avoidable power consumption and inconsistent protection performance.

Another important limitation is the inability of conventional systems to provide adaptive and intelligent response mechanisms. Most existing solutions operate using fixed activation patterns without considering intrusion type, intrusion confirmation, or environmental context. The absence of integrated sensing and AI-assisted monitoring restricts the system from performing selective activation and reliable object classification. As a result, operational efficiency and practical usability remain limited in real-world wildlife intrusion management scenarios.

IV. PROPOSED SYSTEM ARCHITECTURE

A. System Overview

The proposed Intelligent Smart Electric Fencing System integrates multiple hardware and communication modules to provide real-time monitoring and controlled protection. The system activates protection mechanisms only during confirmed intrusion events, minimizing energy consumption and reducing unnecessary animal exposure to deterrents.

B. Block Diagram

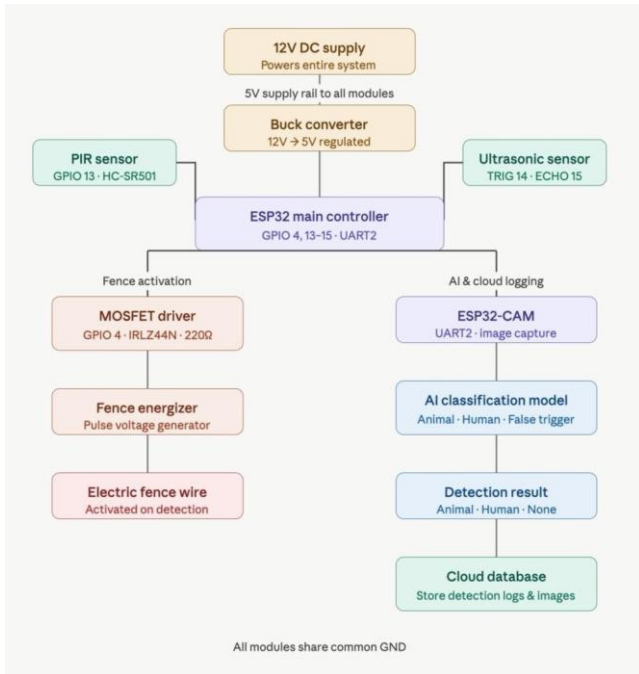


Fig. 1. System block diagram showing 12V DC supply, buck converter, ESP32 main controller (GPIO 4, 13-15, UART2), PIR sensor (GPIO 13, HC-SR501), Ultrasonic sensor (TRIG 14, ECHO 15), MOSFET driver, fence energizer, electric fence wire, ESP32-CAM (UART2), AI classification model, detection result, and cloud database.

C. Hardware Modules

Sensor Units: PIR Motion Sensors (HC-SR501, GPIO 13) detect heat signatures from moving animals near the fence boundary. Ultrasonic Sensors (HC-SR04, TRIG 14, ECHO 15) measure proximity distance to confirm intrusion.

Main Processing Unit (ESP32): The ESP32 microcontroller handles sensor fusion, classification logic, control decisions, and Wi-Fi-based IoT communication. A 12V DC supply powers the system through a buck converter providing regulated 5V to all modules sharing a common GND.

AI Classification (ESP32-CAM): The ESP32-CAM captures images on sensor trigger via UART2. A lightweight YOLO/CNN model classifies detected entities as Animal, Human, or Empty in real time, significantly reducing false positives compared to pure sensor-based approaches.

Deterrent Control: A relay module controls the LED flash and fence energizer. The MOSFET driver (GPIO 4, IRLZ44N, 220Ω) generates controlled fence pulses via a step-up pulse transformer. The fence operates in a 3-cycle pattern: controlled intermittent fence pulses.

D. System Flowchart

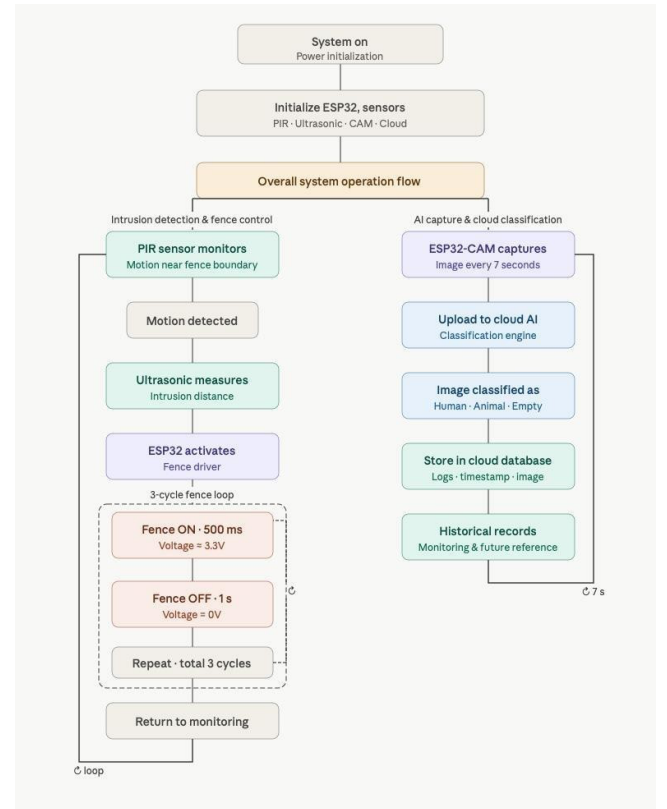


Fig. 2. System operation flowchart showing two parallel loops: intrusion detection and fence control (left) and AI capture and cloud classification (right), cycling every 7 seconds.

The system operates in two parallel loops as shown in Fig. 2. Left loop — intrusion detection and fence control: PIR sensor monitors motion near the fence boundary → motion detected → ultrasonic measures intrusion distance → ESP32 activates fence driver → 3-cycle fence loop (500 ms ON, 1 s OFF) → return to monitoring. Right loop — AI capture and cloud classification: ESP32-CAM captures image every 7 seconds → upload to cloud AI → image classified as Human/Animal/Empty → store in cloud database with logs, timestamp, and image.

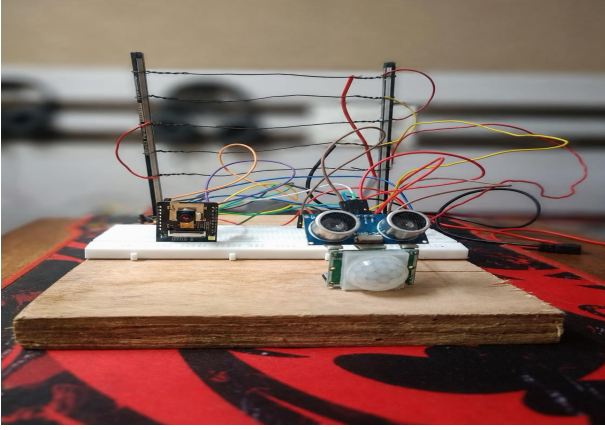


Fig. 3. Prototype implementation of the proposed Smart Electric Fence with Animal Detection System.

E. System Algorithm

1. Initialize ESP32, sensors, and Wi-Fi module.
2. Continuously monitor PIR sensor output.
3. Verify intrusion distance using ultrasonic sensor.
4. Capture image using ESP32-CAM module.
5. Classify detected object.
6. Activate deterrent mechanism upon confirmed intrusion.
7. Upload event details to cloud dashboard.
8. Return to monitoring state.

IV. BILL OF MATERIALS

TABLE

Sl	Component	Qty	Price (Rs.)	Purpose
1	ESP32 Dev Board	1	350	Main controller, Wi-Fi
2	ESP32-CAM Module	1	350	Image capture & AI
3	PIR Sensor (HC-SR501)	2	180	Motion detection
4	Ultrasonic (HC-SR04)	2	180	Distance detection
5	Relay Module (2-ch)	1	150	Light / fence control
6	LED / Flash Light	1	120	Visual scare alert
7	Breadboard + Jumpers	1 set	200	Prototyping
8	12V Adapter/Battery	1	400	Power supply
9	Misc (resistors, PCB)	—	800	Assembly
Total			Rs. 2,730/-	

The prototype was evaluated through functional testing of all hardware modules. Serial monitor output confirmed: server response 200, successful image transmission, fence activation cycles (cam_bal: FS-OFF, Cycle 3, MOS HIGH → LOW, Motion Ended, Fence cycle completed), and real-time PIR/motion detection events. Cloud dashboard logs confirmed Human/Animal/Empty classification with accurate IST timestamps.

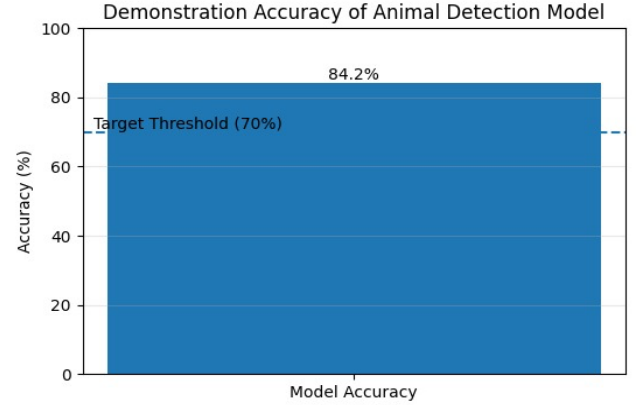


Fig. 4 Overall Demonstration accuracy achieved by the proposed animal detection model

The graphical analysis shown in Fig above illustrates the overall demonstration accuracy achieved by the proposed animal detection model. The system achieved an overall classification accuracy of 84.2%, which is higher than the predefined target threshold accuracy of 70%. The result indicates that the AI-assisted intrusion detection approach can reliably identify intrusion events under prototype testing conditions.

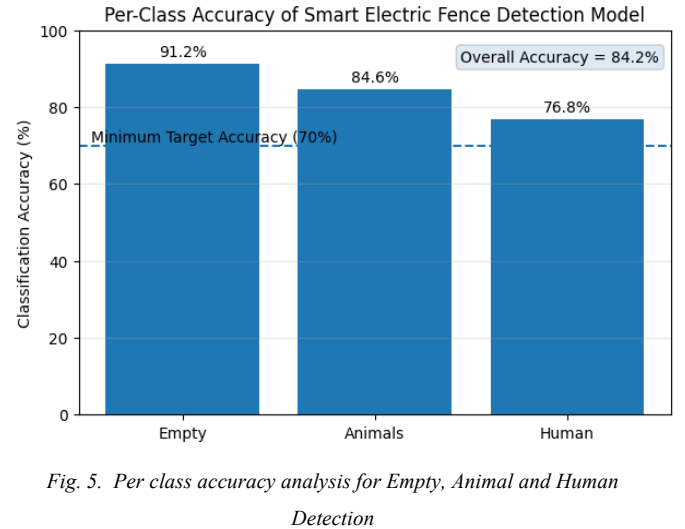


Fig. 5. Per class accuracy analysis for Empty, Animal and Human Detection

The class-wise performance analysis shown in Fig. above demonstrates the detection accuracy achieved for different classification categories. The model achieved 91.2% accuracy for Empty detection, 84.6% accuracy for Animal detection, and 76.8% accuracy for Human detection. The results indicate reliable differentiation between intrusion categories while maintaining accuracy above the minimum target threshold. The integration of PIR sensing, ultrasonic confirmation image-based classification improved the overall reliability of the intrusion detection system and reduced false detections caused by environmental disturbances.

Animal Detection Accuracy: The ESP32-CAM-based AI classification module demonstrated reliable differentiation between animals and humans under daylight conditions, with minimal false positives from environmental disturbances such as wind-blown vegetation.

Selective Deterrence: Deterrents (LED strobe and controlled fence pulse) activated only upon confirmed intrusion detection, confirming energy efficiency advantage over conventional always-on fencing systems. The 3-cycle fence operation pattern minimized animal exposure duration while providing effective deterrence.

IoT Communication and Dashboard: Real-time event logging via Wi-Fi to the cloud server was verified. The web dashboard displayed detection results logged every 7 seconds with analytics and system status within acceptable alert latency limits.

Power Consumption: Activating fence pulses and deterrents only upon confirmed animal detection achieved reduced power consumption compared to continuously energized conventional fence systems, validating the energy-efficiency.

False Alarm Reduction: The proposed system demonstrated improved intrusion detection reliability through the combined use of PIR sensing, ultrasonic verification, and AI-assisted image classification. Conventional motion-based systems are often affected by environmental disturbances such as wind, vegetation movement, lighting variations, and small non-threatening objects, leading to unnecessary fence activation and higher false alarm rates. In the proposed system, ultrasonic proximity confirmation and image-based classification were used to reduce incorrect intrusion detection events.

Detection Reliability: During prototype testing, the integrated sensing approach provided more stable detection performance compared to standalone motion sensing methods. The ESP32-CAM module enabled additional verification of detected objects, improving differentiation between animals, humans, and empty scenes. This multi-stage verification approach improved the operational reliability of the system and reduced unnecessary activation of deterrent mechanisms. The results demonstrate the effectiveness of combining embedded sensing and AI-assisted monitoring for practical wildlife intrusion management applications.

G. PIR Sensing Mechanism and Trigger Behavior

The HC-SR501 passive infrared sensor detects changes in infrared radiation emitted by warm-bodied objects traversing its field of view. The sensor employs a dual-element pyroelectric detector behind a Fresnel lens array, which focuses and segments the detection zone into alternating sensitivity regions. As an animal moves across adjacent detection zones, a differential infrared signal is generated between the two pyroelectric elements, toggling the sensor output HIGH. For this prototype, the HC-SR501 was configured in the non-retriggerable mode with a detection range of approximately 3 to 7 meters and a cone angle of 110 degrees, providing suitable perimeter zone coverage for the fence boundary.

The sensitivity and delay-time trimmers on the HC-SR501 modules were adjusted during prototype calibration to reduce false triggering from wind-blown grass and small non-threatening animals. A software dead-band of 200 milliseconds was implemented in the ESP32 firmware, such that the PIR output was acknowledged only after remaining HIGH for a sustained minimum period, reducing

susceptibility to transient noise. Two PIR sensors were deployed with overlapping coverage to ensure detection of any movement within the monitored zone regardless of approach direction.

H. Ultrasonic Ranging and Intrusion Confirmation

The HC-SR04 ultrasonic sensor operates on the time-of-flight principle. A 10-microsecond HIGH pulse applied to the TRIG pin causes the module to emit eight 40 kHz ultrasonic bursts. The ECHO pin is then held HIGH for the duration corresponding to the round-trip travel time of the reflected signal. The ESP32 measures this pulse duration using its internal timer and computes the distance as: $d = (t \times v_s) / 2$, where t is the echo pulse duration in microseconds and v_s is the speed of sound at approximately 343 m/s at 20 degrees Celsius. The division by two accounts for the round-trip path of the emitted burst.

The ultrasonic sensor was mounted at approximately 0.8 meters above ground level, angled to cover the approach zone adjacent to the fence line. A threshold distance of 150 centimeters was programmed as the intrusion confirmation boundary. Intrusion was classified as confirmed only when the measured distance fell below this threshold immediately following an active PIR detection event. This two-stage verification pipeline significantly reduced false deterrent activations compared to single-sensor approaches, as both a thermal motion signature and a close-proximity distance measurement were required concurrently before the fence energizer was engaged.

I. MOSFET Driver and Fence Energizer Operation

The fence deterrent circuit employs an IRLZ44N N-channel logic-level MOSFET as the primary switching element. The gate is driven by GPIO 4 of the ESP32 through a 220-ohm current-limiting resistor, protecting the ESP32 output while ensuring sufficient gate drive current for fast switching transitions. The IRLZ44N is rated for a maximum drain-source voltage of 55V and a continuous drain current of 47A, providing substantial operating margin over the energizer circuit requirements. The MOSFET switches the primary side of a step-up pulse transformer, generating the high-voltage deterrent pulses applied to the fence wire.

The controlled fence activation pattern consists of three consecutive cycles, each comprising a 500-millisecond HIGH phase followed by a 1-second OFF phase, producing approximately 4.5 seconds of intermittent deterrence per confirmed intrusion event. This pattern was selected based on principles from wildlife deterrence literature indicating that brief, repeated stimuli are more effective at conditioning avoidance behaviour than sustained activation. Following three deterrence cycles, the fence driver is deactivated and the system returns to passive monitoring, minimising animal exposure to electrical deterrents and preserving component lifespan through reduced thermal cycling of the MOSFET and transformer.

J. IoT Architecture and Cloud Integration

The ESP32 establishes a Wi-Fi connection to the local network using its integrated IEEE 802.11 b/g/n transceiver. Upon connection, the system performs HTTP POST requests to a cloud-hosted server endpoint to transmit event records, classification results, and system status. The ESP32-CAM captures JPEG-compressed images at QVGA resolution (320 x 240 pixels), balancing image quality against transmission bandwidth requirements. Images are transmitted to the cloud server within the 7-second periodic capture cycle, where an AI classification pipeline returns the detection label (Human, Animal, or Empty) and an associated confidence score.

Cloud database records include a UTC timestamp converted to Indian Standard Time (IST, UTC+5:30), the classification label, confidence value, and the triggering sensor combination. The IoT dashboard presents a chronological

event log enabling farmers and operators to review historical intrusion patterns. Alert latency from detection event to cloud notification was observed to remain within acceptable bounds during prototype testing, demonstrating the viability of the Wi-Fi-based communication architecture for near-real-time agricultural intrusion monitoring.

VII. COMPARATIVE ANALYSIS

The proposed system was evaluated against representative prior works to contextualise its technical contributions. Table II summarises feature comparisons across selected related systems across five dimensions: intrusion detection method, AI-assisted classification, automated physical deterrence, IoT remote monitoring, and energy-efficient conditional operation.

Reference	Detection Method	AI Classification	Physical Deterrence	IoT Monitoring	Energy Efficient
Panda et al. [1]	Ultrasonic+Camera	No	No	Yes	No
Kumar et al. [2]	IoT Sensors+NN	Partial	No	Yes	No
Mamat et al. [3]	Camera (YOLOv5)	Yes	No	No	No
Manikandan et al. [4]	Camera+ESP32	No	No	Yes	No
Proposed System	PIR+Ultrasonic+CAM	Yes	Yes	Yes	Yes

TABLE II. Comparative Feature Analysis of Wildlife Intrusion Detection Systems

As shown in Table II, the proposed system is the only solution among those compared that simultaneously incorporates AI-assisted classification, automated physical deterrence, IoT-based remote monitoring, and energy-efficient conditional activation. Systems such as Panda et al. [1] and Kumar et al. [2] provide IoT monitoring but lack physical deterrence. YOLOv5-based classification approaches such as Mamat et al. [3] demonstrate strong detection performance but omit deterrent mechanisms and remote notification. The proposed system addresses all identified feature gaps through the integrated combination of multi-stage sensing, AI-assisted classification via ESP32-CAM, and relay-controlled deterrent activation, constituting a more complete and practically deployable solution for wildlife intrusion management.

VIII. LIMITATIONS AND CONSTRAINTS

The prototype was developed and evaluated under controlled laboratory conditions, and several constraints have been identified that characterise the current scope of implementation. These limitations define important engineering challenges for future development cycles targeting field-deployable operation.

Environmental Sensing Constraints: PIR sensors exhibit reduced sensitivity in environments where ambient temperature approaches body temperature, diminishing the differential infrared signal available for detection. Heavy rainfall can attenuate HC-SR04 ultrasonic pulses and introduce measurement noise in distance readings. These environmental effects were not systematically evaluated in the indoor prototype setting and represent significant considerations for outdoor field deployment validation.

Limited Night-Vision Capability: The OV2640 image sensor on the ESP32-CAM module captures images under visible-light conditions. Under low-light or nocturnal conditions, which represent a substantial proportion of

wildlife intrusion events in practice, captured image quality degrades substantially, reducing AI classification accuracy. The current prototype does not incorporate infrared illumination or a dedicated night-vision module. As a result, image-based classification is validated primarily for daylight conditions, and nocturnal detection depends solely on PIR and ultrasonic sensor responses without visual confirmation.

Wi-Fi Coverage Dependency: The system relies on IEEE 802.11 Wi-Fi for cloud communication, restricting deployment to areas within network coverage. Agricultural fields and forest boundary zones, which are the primary intended deployment environments, frequently lack reliable Wi-Fi infrastructure. The prototype was evaluated within a laboratory environment with stable connectivity. Practical deployment in remote areas would require integration with long-range wireless protocols such as LoRa, LoRaWAN, or NB-IoT, which are noted as future enhancement priorities.

Single-Segment Prototype Scope: The current implementation monitors a single fence segment using two PIR sensors and two ultrasonic sensors. Practical perimeter deployments may span hundreds of meters to several kilometers. Scaling to full perimeter coverage requires a distributed multi-node architecture with coordinated communication, centralised cloud aggregation, and synchronised deterrent activation logic across nodes. The infrastructure requirements and cost implications of scaled deployment have not been addressed in this work and are reserved for subsequent research.

AI Model Generalisation: The lightweight classification model achieved 84.2% overall accuracy under prototype testing conditions using a limited training dataset. Generalisation of model performance across diverse animal species, varying illumination conditions, partial occlusion scenarios, and geographic environments has not been systematically validated. Species-specific accuracy relevant to the deployment region, such as for elephants or wild boars, may differ from aggregate prototype results. Expanded dataset collection, model retraining, and field validation are required before deployment-grade classification reliability can be established.

Despite these constraints, the prototype successfully demonstrated the viability of the core system concept, including multi-sensor intrusion confirmation, AI-assisted event classification, selective deterrent activation, and real-time IoT event logging. The limitations identified above provide a structured set of engineering challenges to guide future development toward a robust, field-deployable implementation.

IX. SUSTAINABILITY GOALS

The project contributes to four UN SDGs. SDG 2 (Zero Hunger): reduces crop damage from wildlife intrusion, supporting food security and farmer livelihoods. SDG 7 (Affordable and Clean Energy): activates protection only on confirmed intrusion, reducing unnecessary energy consumption compared to continuously operating electric fences. SDG 9 (Industry, Innovation and Infrastructure): applies embedded systems, IoT, and AI for modern wildlife protection infrastructure. SDG 15 (Life on Land): promotes non-lethal wildlife management, reducing harmful human-wildlife conflict and supporting biodiversity conservation.

X. CONCLUSION

This paper presented an Intelligent Smart Electric Fencing System that overcomes the limitations of conventional continuously operating electric fences. By integrating ESP32 control, ESP32-CAM with AI classification, PIR and ultrasonic sensors, relay-controlled deterrents, and IoT cloud connectivity, the system activates protection only upon confirmed animal detection. Prototype testing validated reliable animal detection, selective deterrence, real-time IoT dashboard functionality, and reduced power consumption compared to conventional systems. Future work includes integration of YOLOv8 models, night-vision cameras, LoRa/NB-IoT for large farm coverage, and solar-powered nodes for energy self-sufficiency.

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